

Sustainability notes

Me

Today

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1 1 - What is sustainability?

Sustainability is about ensuring well-being equitability within planetary limits and ensuring that this balance is maintained for future generations.

key dimensions to consider when appraising sustainability

1. Environmental
2. Economical
3. Social

The S.M.A.R.T objective

An objective that is specific, measurable, achievable, relevant and time-bound.

2 2 - Life cycle approach

The I=PAT equation

Impact = People Wealth Technologies

$$I = P A T = \text{Pop} \cdot \frac{\text{GDP}}{\text{Person}} \cdot \frac{I}{\text{GDP}}$$

1. I is the environmental impact
2. Pop is the global population
3. $\frac{\text{GDP}}{\text{Person}}$ is the Affluence, the material standard of living
4. $\frac{I}{\text{GDP}}$ Is the technology factor

Where the technology factor needs to decrease by a factor of 4-20 to compensate for increasing P and A and an already large I.

The **rebound effect** occurs because technological improvements often will reduce the price of the products, making money accessible to buy more of the same product or buy other products. This may cause an increase in environmental impact, or reduce the environmental improvement. Fair wage is one of the most applicable

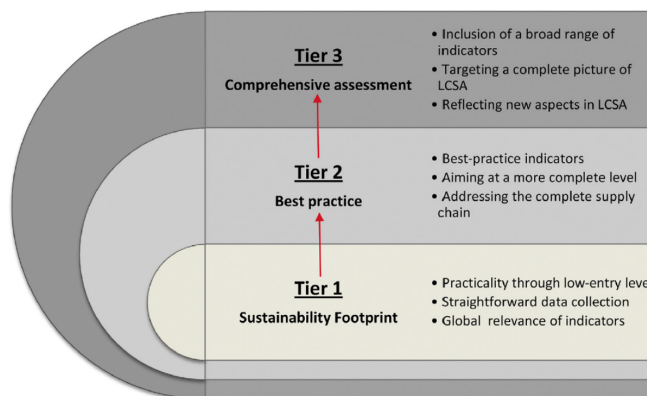


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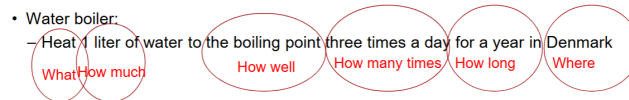
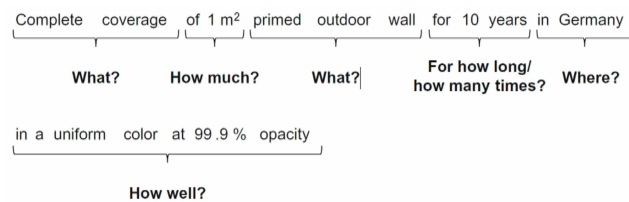
indicators considered within SLCA. It also describes perfectly a necessity to fulfil basic needs. Therefore it is the first level indicator

Health has been broadly addressed within SLCA and LCA literature and by numerous institutions, like WHO or OECD. Therefore, they have considered it a tier 2 indicator.

Human rights are normally expressed by different subtopics, e.g. child labour, forced labour, discrimination and equity and therefore it is a more complex indicator to be considered at level 3.

Functional unit:

Focus on the service that the product, system or technology provides

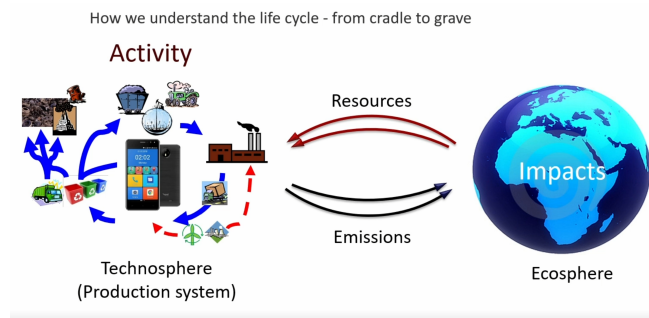


3 3 - Case study

Just keeping this st. numbers match :D

3.0.1 The assesment framework

LCA - Life cycle assement



- Definition of goal of assessment - "what is the question"
- Scoping the system
- Collection of data on emission and resource use
- Translation of emission into enviromental impact



4 4 - Carbon footprint and Environmental indicators

- **Primary energy** is defined as the energy contained in the energy carriers at the point of extraction from the environment, including the upstream energy usage for extraction, preparation, and distribution.
- **final energy** is the energy purchased by the consumer and delivered to him.
- **useful energy** is the portion of final energy which is actually available after final conversion to the consumer for the respective use

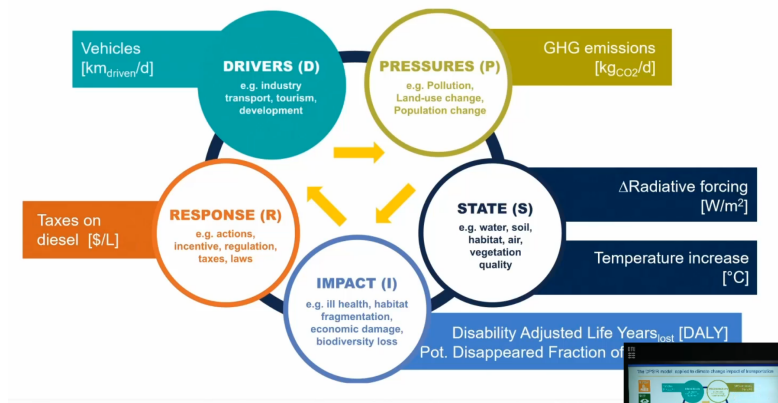
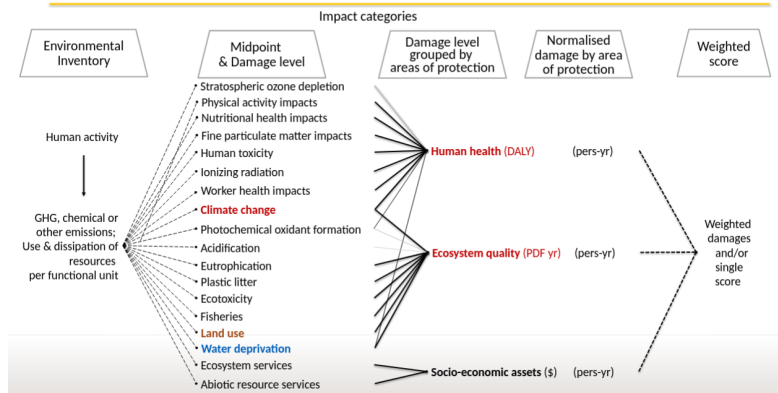


Figure 2: DPSIR model - Applied to impact of transport

image

UNEP - GLAM: the global life cycle impact assessment framework



Greenhouse Gas	GWP100
Carbon Dioxide (CO ₂)	1 kg _{CO2e} /kg _{CO2}
Methane biogenic (CH ₄)	27 kg _{CO2e} /kg _{CH4}
Methane fossil (CH ₄)	29.8 kg _{CO2e} /kg _{CH4}
Nitrous Oxide (N ₂ O)	273 kg _{CO2e} /kg _{N2O}
Hydrofluorocarbons (HFC-32)	771 kg _{CO2e} /kg _{HFC}
Perfluorocarbons (PFC-14)	7380 kg _{CO2e} /kg _{PFC}
Sulphur Hexafluoride (SF ₆)	24300 kg _{CO2e} /kg _{SF6}

5 5 - Economics dimension

6 6 - Planetary boundaries and absolute sustainability

6.0.1 Eco-efficiency

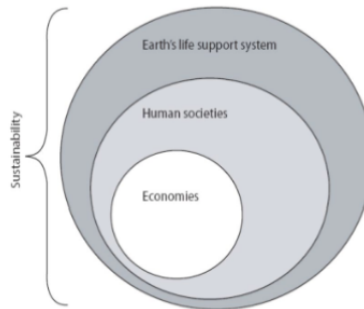
Defined as the ratio between functional output and environmental impact caused by an activity

$$\text{Eco-efficiency} = \frac{\text{Delivered service}}{\text{Environmental impact}} = \frac{1}{T}$$

It's the reciprocal of the technology factor and are determined with life cycle assessment.

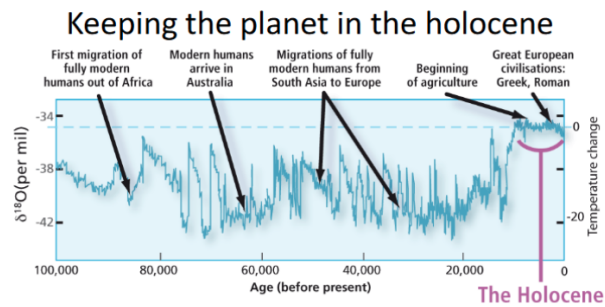
6.0.2 The three dimensions of sustainability

The idea behind **absolute sustainability** is to take a top down approach. Meaning that we prioritize earth and society before we prioritize economics.



6.0.3 The planetary boundary approach

There are some uncertainty to why we have been at a stable temperature for the last 10.000 years, but our best guess is the 9 following parameters:



6.0.4 Aesa

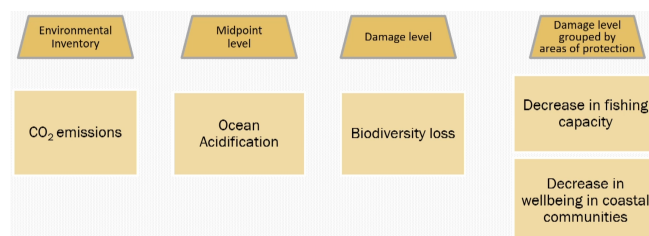
1. What are the biophysical boundaries and determine safe operating space.
2. Allocating the operating space.
3. Calculate the impact profile of the activity using LCA
4. Compare impact and allocated space. (If it's smaller it's sustainable)

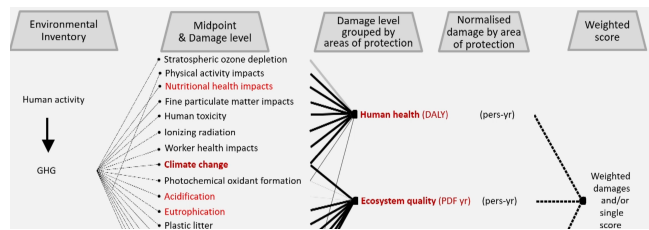
7 7 - Social and health

7.0.1 SLCA

S-LCA four main steps

- Objectives and scope definitions
- Determine stakeholder and associated inventory indicators
- social impacts assessments
- Interpretation





7.0.2 The Global Burden of Disease (GBD)

Measure - "Disability adjusted life years

- 1 DALY = 1 year of healthy life lost
- DALY = YLD (Years with disability) - YYL (Years of life lost)

We also have a disability weight so 1 year with cancer gives a YLD of 0.3.

7.0.3 Exposome definition

The exposome is composed of every exposure to which an individual is subjected from conception to death. Therefore, it requires consideration of both the nature of those exposures and their changes over time. For ease of description, three broad categories of non-genetic exposures may be considered: internal, specific external and general external.

The exposome includes all exposures from conception to death:

- **General external:** social status, education, climate.
- **Specific external:** pollution, diet, chemicals, lifestyle.
- **Internal:** metabolism, physical activity, aging.

Health = function of genome + exposome.

Enables comparison of impacts from pollution, food, chemicals, exercise, and climate change in a common health metric.

7.1 Major exposure insight

- Air pollution (PM2.5) causes 6.5 million deaths/year globally.
- Indoor sources (e.g., candles) can cause measurable minutes of healthy life lost per hour.
- Consumer product chemicals can lead to 20–1500 minutes of healthy life lost per user per day for certain product–chemical combinations.
- Nutrition impacts quantified via the Health Nutritional Index (HENI) (e.g., one beef hot dog $\approx$ 36 minutes of healthy life lost).
- Physical activity yields substantial health gains (walking, running, cycling).

Eating a single hot dog could take 36 minutes of your life, a new study says.

8 8 - Resource Management, Scarcity and Criticality

8.0.1 Fundamentals

"Degrading common resources affect the ability for others to extract benefit of them"

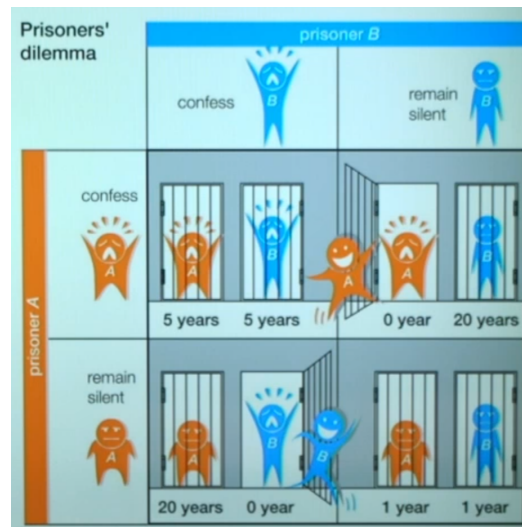
- Excusable - Property rights can be assigned
- Rivalrous - Use of that good prevents others from using it.

	Excludable	Non-excludable
Rivalrous	Private good	Common good
Non-rivalrous	Club good	Public good

- Private good - Parking space, bread a field
- Club good - Cinema, club or private park
- Public good - Knowledge, streetlights, public park
- Common good - Fish stocks and public parks

8.0.2 Tragedy of the commons

The best strategy is to confess, because you can't control what others are doing. This is like a fish stock, you have to maximize fishing st. others don't. Since only 1 country get the gain, all share the cost. The solution is



to assign property rights st. it becomes a private good.

8.0.3 Finding sustainable trade-offs

repopulation formula for fish

$$B_{t+1} = B_t + rB_t \left(1 - \frac{B_t}{K}\right) - C_t$$

MSY $\Rightarrow$ Maximum sustainable yield

MEY $\Rightarrow$ Maximum economic yield

9 9 - Applying Tools

10 10. Circular economy

All 4 sustainability areas effect each other.

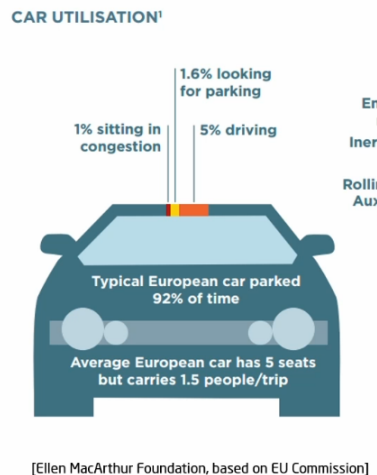
- Global warming
- Toxic discharge
- Ressource
- Biodiversity

10.1 What is the problem?

10.1.1 The problem with linear economy

Take → make → use → dispose

Structural waste Only 5% used for driving **Rebound effect**

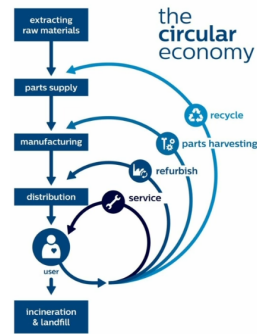


There is no such thing as throwing away, the material will always need to be somewhere. That's way we need to recycle.

10.2 The promise of circularity

Close loops, how can we close the loops of resources. St. there are no waste, but everything is recycled.

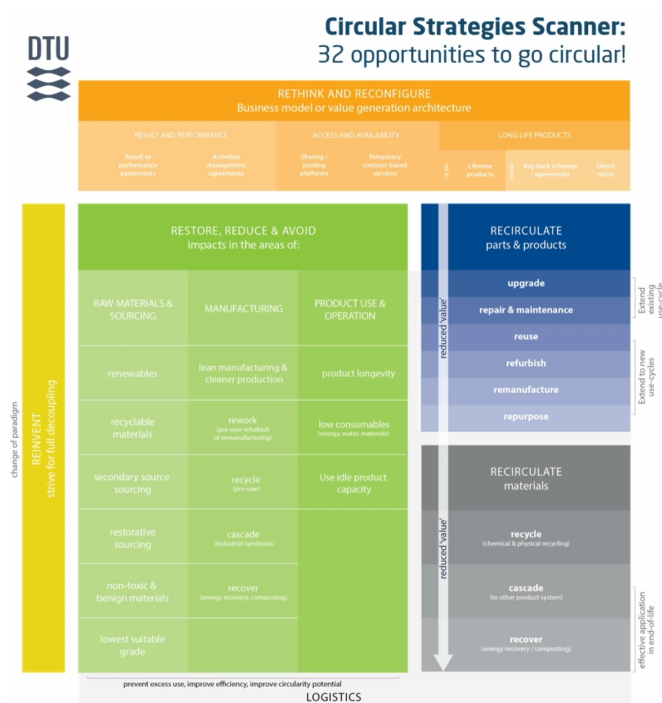
Decouple value creation from resource consumption. How can we keep fulfilling our needs as a user without increasing resource demand.



10.3 Strategies

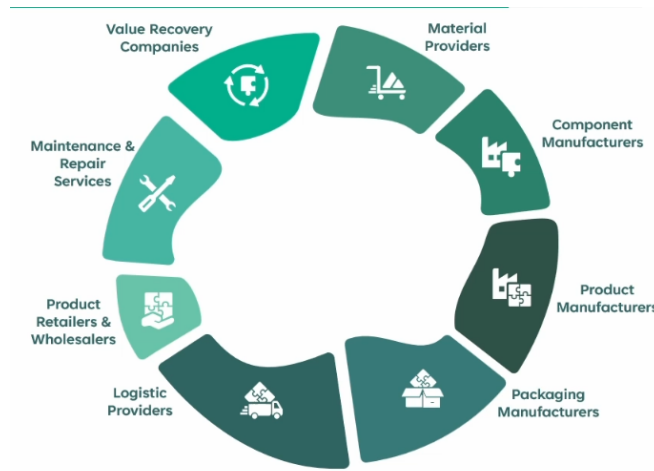
Recirculate - Material

- Recycle - Use materials for the same product again.
- Cascade - Use materials in other systems
- Recover - Energy recover / Composition

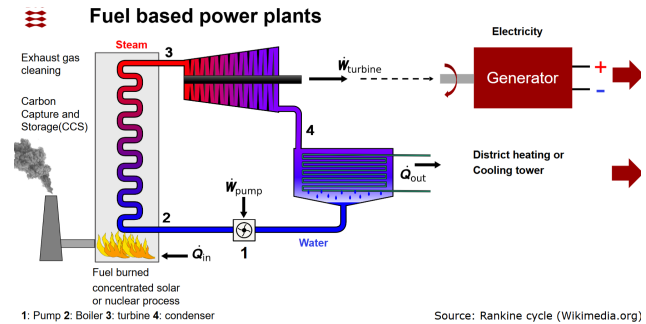


10.4 Readiness

You need to be able to collaborate in the value chain.



11 11 - Energy Systems and Transition/Decarbonization



• Electricity Supply – National Governments

- Remove fossil fuel subsidies in a socially acceptable manner
- Remove barriers to the expansion of renewables
- Stop expansion of fossil fuel infrastructure
- Plan for a just fossil fuel phase-out
- Adapt electricity market rules for high shares of renewables

• International Cooperation

- Cooperate on a just coal phase-out
- Support initiatives on emissions-free electricity, power system flexibility, and interconnection solutions

• Subnational Governments

- Set 100% renewable electricity targets
- Plan for a just fossil fuel phase-out

• Businesses

- Support a 100% renewable electricity future

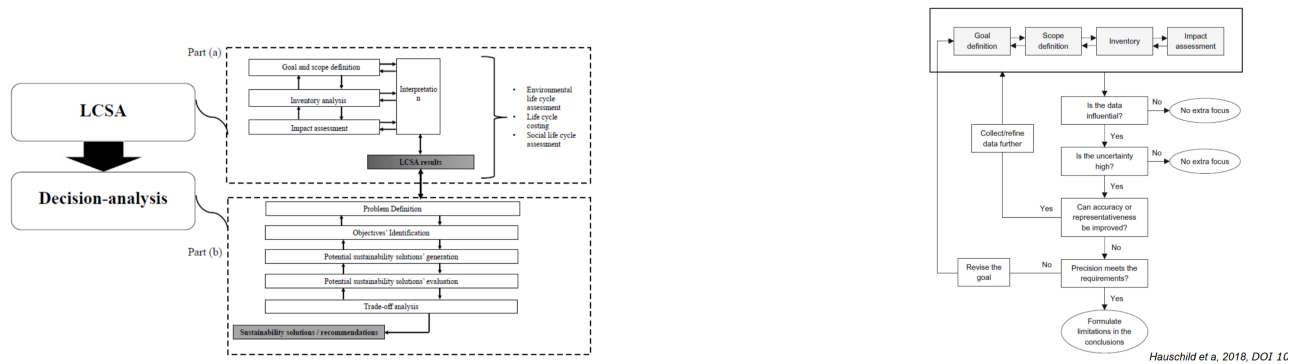
• Investors, Private and Development Banks

- Engage with or divest from fossil fuel electricity utility companies
- Do not invest in or insure new fossil fuel infrastructure

• Citizens

- Purchase 100% renewable electricity

12 12 - Sustainable Transition



12.1 Power plant emmsion

See slide 581

